

Unit 5

INTERMEDIATE SQL

1. INDEX IN SQL

Definition

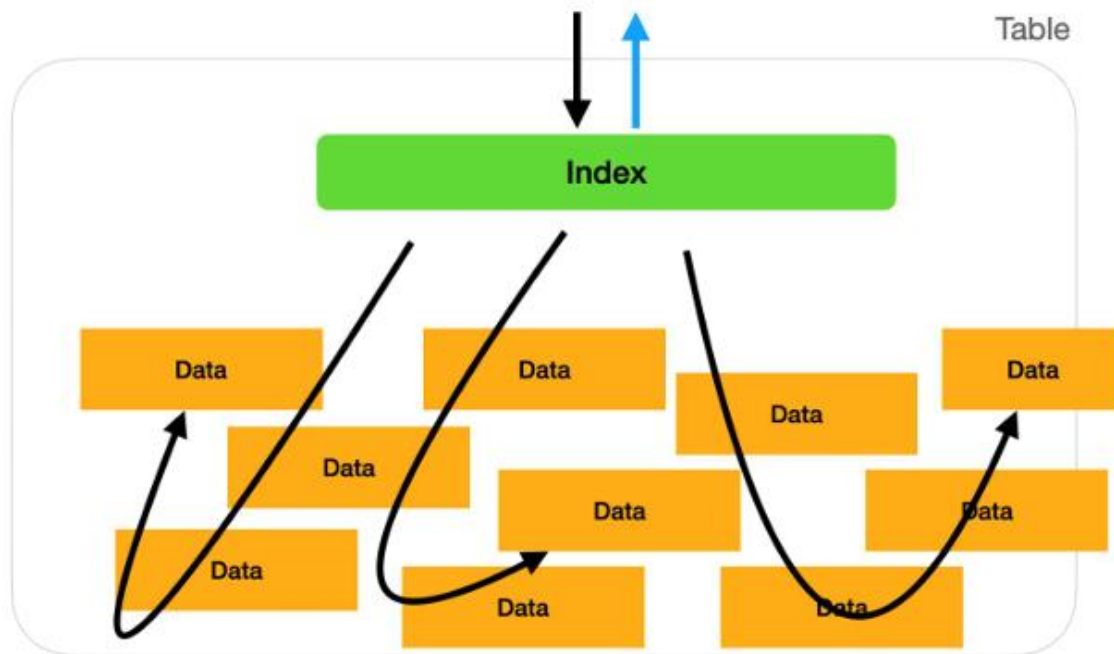
- An index is used to find data in a table quickly. It improves the performance of SQL queries and reduces searching time.

Understand Index

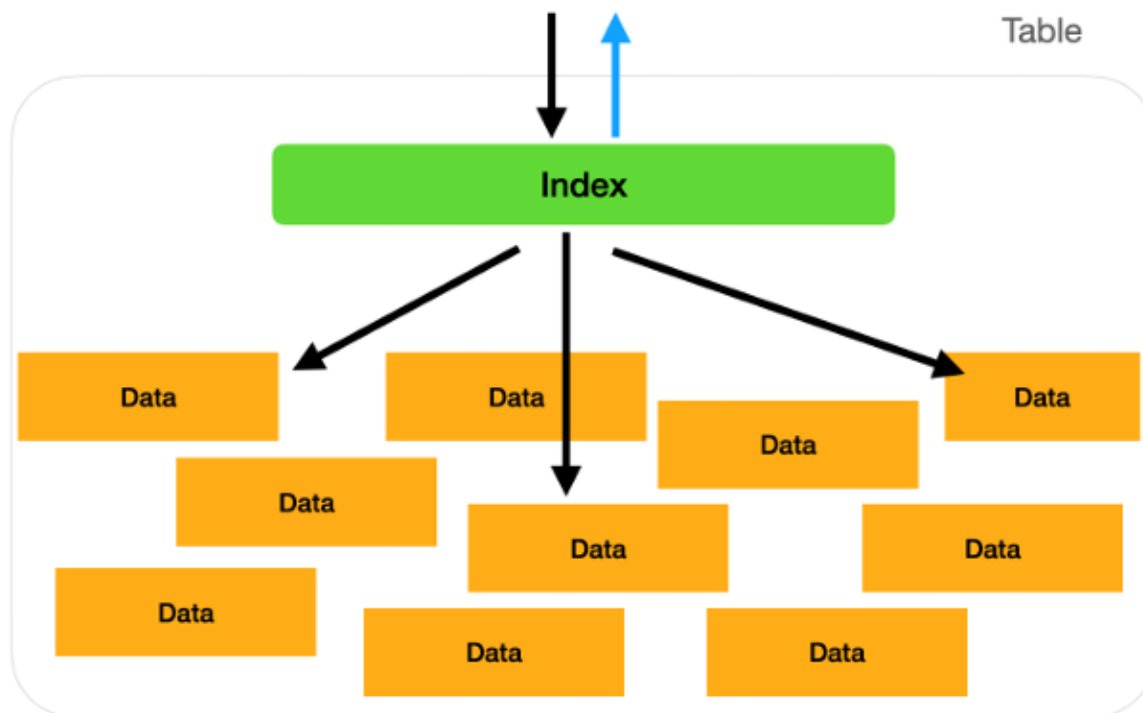
Concept & Pros

- The core concept of index technology is to mark specific columns in the table and extract them to be stored additionally and in an organized manner.

Without INDEX: In the index layer, no index is found, so many redundant query paths may be taken to retrieve data.



With index: In the index layer, if an index is found and it has relative information to



Advantages of Index

1. Fast Searching: Helps to find data quickly.
2. Improves Performance: Makes SQL queries faster.
3. Faster JOIN: Helps to combine tables efficiently.

Disadvantages of Index

1. It requires additional disk space.
2. Frequent INSERT and UPDATE operations may become slower because the index must be maintained.
3. Indexes are not necessary for every column.

Syntax

```
CREATE INDEX index_name  
ON table_name (column_name);
```

Example:

```
CREATE INDEX idx_name  
ON Student (name);
```

This creates an index on the name column of the Student table.

To delete an index:

```
DROP INDEX idx_name ON Student;
```

Types of Index

1. PRIMARY KEY: Uniquely identifies each row and automatically creates an index.
2. UNIQUE INDEX: Prevents duplicate values in the indexed column.
3. FULLTEXT INDEX: Helps to search for keywords in text.
4. INDEX (KEY): Helps to retrieve data quickly.

Six Principles About Index Best Practices

When using an index, we recommend mastering these six principles so that the index can play a high-quality role in our code.

1. Try to choose UNIQUE INDEX, otherwise, use KEY.
2. Add an index for the column that is often referenced by complex operations, such as GROUP BY.
3. Index the hotspot column that is frequently queried in WHERE, but make sure that the differences in the column values are large enough. For example, adding an index to the identify_number column is better than adding an index to country_of_birth.
4. The index actually uses the disk space, so pay attention to the space used in the hard disk.
5. Index the column with a small value. For example, the index title of the article is better than the index content of the article .
6. Try to avoid index on columns with NULL, which may affect the query efficiency.

The above is the syntax of the index and the 6 basic principles that need to be mastered. I hope it can help you use the index more correctly!

2.SQL USER MANAGEMENT

Definition:

SQL User Management creates users and controls database permissions.

1.Create User: Creates a new user.

```
CREATE USER 'john'@'localhost' IDENTIFIED BY 'password';
```

2.Network Context:

- "localhost" – Same computer.
- "IP Address" – Specific computer.
- "%" – Any host.

3.Show Users: Displays user details.

```
SELECT User, Host FROM mysql.user;
```

4.Grant Permissions: Gives permissions to users.

```
GRANT SELECT ON college.Student TO 'john'@'localhost';
```

Conclusion

SQL indexes improve data searching and query performance. SQL User Management helps create users and control database access. Using indexes correctly and assigning suitable permissions improves database efficiency and security.